Start here

AI and Data Science Professional | KLAi Professional Learning

Welcome. This course takes you from the spreadsheet you already know to a tested model that a colleague can open. You do not need to code or to know statistics. You need to be comfortable with Excel or Google Sheets. There are 24 sessions of 2 hours, 48 hours in all, and every tool is free.

How a session runs

Minutes	What happens
10	Recall three questions from last time, and meet the question of the day
45	Ideas first: pictures, plain words and one example by hand. Laptops closed.
10	Break
45	Lab: you predict, run, read and change, in a Google Sheet or a Colab notebook
10	Exit check, what you keep and what comes next

The ideas come before the code. After the break a black slide says Close the slides. Open the notebook. From then on you work on your own laptop and your tutor walks round.

What to bring

- A laptop (Windows or Mac) with a browser and a charger. A phone is too small for the lab.
- A free Google account.
- A notebook and a pen.
- One question from your own work that you would like data to answer. You use it in Session 1 and again for your course project.

The free tools

- **Google Colab.** Runs Python in your browser. Nothing to install.
- **Google Sheets** (or Excel). Sessions 1 to 4, Session 24 and every comparison with Python.
- **Python libraries:** pandas, NumPy, SciPy, statsmodels, scikit-learn, matplotlib and seaborn. They are already in Colab.
- **SQLite** for SQL, inside Colab.
- **GitHub and Streamlit Community Cloud** to save your work and publish a small app (Sessions 23 and 24).
- **TensorFlow Playground** in your browser, for neural networks (Session 20).

Free plans and their limits change. Your tutor tells you if something has moved. The course data are fictional and generated. Never put real customer or patient data into a notebook, a chat assistant or an app.

Set up before Session 1 (about 20 minutes)

- **1. A Google account.** If you have Gmail you have one. Otherwise make one at accounts.google.com.
- **2. Open Colab once.** Go to colab.research.google.com and sign in. Choose New notebook, type print("Hello from Colab") in the first cell and press the play button. Then choose File, Save a copy in Drive.
- **3. The workbook.** Your tutor gives you lakeside_for_sheets.xlsx. Upload it to Google Drive, open it and choose File, Save as Google Sheets.
- **4. The data file.** Your tutor gives you a Google Drive link to data.zip, or the file itself. Each notebook starts with a setup cell that downloads it from the link (the line DATA_LINK) or asks you to upload it, and unpacks it.

How the checks work

Every notebook has check cells. A check compares your answer with the right one and prints OK or NOT YET. NOT YET means your answer differs. Read the step above it, change your code and run it again. You may try as often as you like. At the end a line tells you how many checks pass. The checks show you that the lab worked. They are not marks, and nobody marks them. There are no exams in this course. You build projects and keep them. In Sessions 1 to 4 and 24 you work in Google Sheets and compare your answers with the check table on the lab slide. Session 1 also has a short notebook with checks.

Each step has the rhythm PREDICT, RUN, READ, CHANGE. Write a guess before you run a cell. Run it. Say what the output means. Then change one thing and see what happens. Before you finish, choose Runtime, Restart session and run all, so that the notebook works from the top.

How to revise

- Your tutor shares the slides as a PDF. Read it again after class. Every term is defined where it first appears and every picture has a caption, so the slides make sense without the speaker.
- The Glossary lists every term of the course, with its plain meaning and the session where you meet it.
- Words you will meet at work come with the proper industry name first and the plain explanation beside it. Learn the industry name.
- Your saved notebooks are your own reference. Add a comment wherever you learn something.

What you keep from each module

Module	Sessions	You keep
A Think with data	1 to 4	A Google Sheet with your data dictionary, pivot tables, a simulation and your own business question.
B Python for data	5 to 9	Notebooks that load, clean and reshape a file, a cleaning log, and a cleaned file you can reuse.
C Analyse and explain	10 to 13	A SQL query book, an exploration notebook, an inference notebook and a one page regression report.
D Machine learning	14 to 19	A baseline and models for loan default, a customer segment table, and a tested forecast.
E Neural networks and text	20 to 22	A network result table, a scored feedback file and a small search over documents with a test sheet.
F Ship it and present	23 to 24	A folder with your model, requirements and app, a model card, and your course project presentation.

How to ask for help

- Colab will not open: try Chrome or Edge and turn off any extension that blocks pages.
- A cell shows red text: read the last line first. It says what went wrong. Session 5 practises this.
- You have no Google account and do not want one: tell your tutor before Session 1.
- You are stuck in the lab: ask your neighbour for two minutes, then ask your tutor. Write down any question you could not ask and bring it to the next session.
- You missed a session: read the PDF slides and run the notebook before the next one.